

Banking Dataset - Marketing Targets

Exploratory Data Analysis for Bank Marketing

<https://www.kaggle.com/datasets/prakharrathi25/banking-dataset-marketing-targets>

Dataset Overview

This dataset consists of **45,211 rows** and **17 columns**

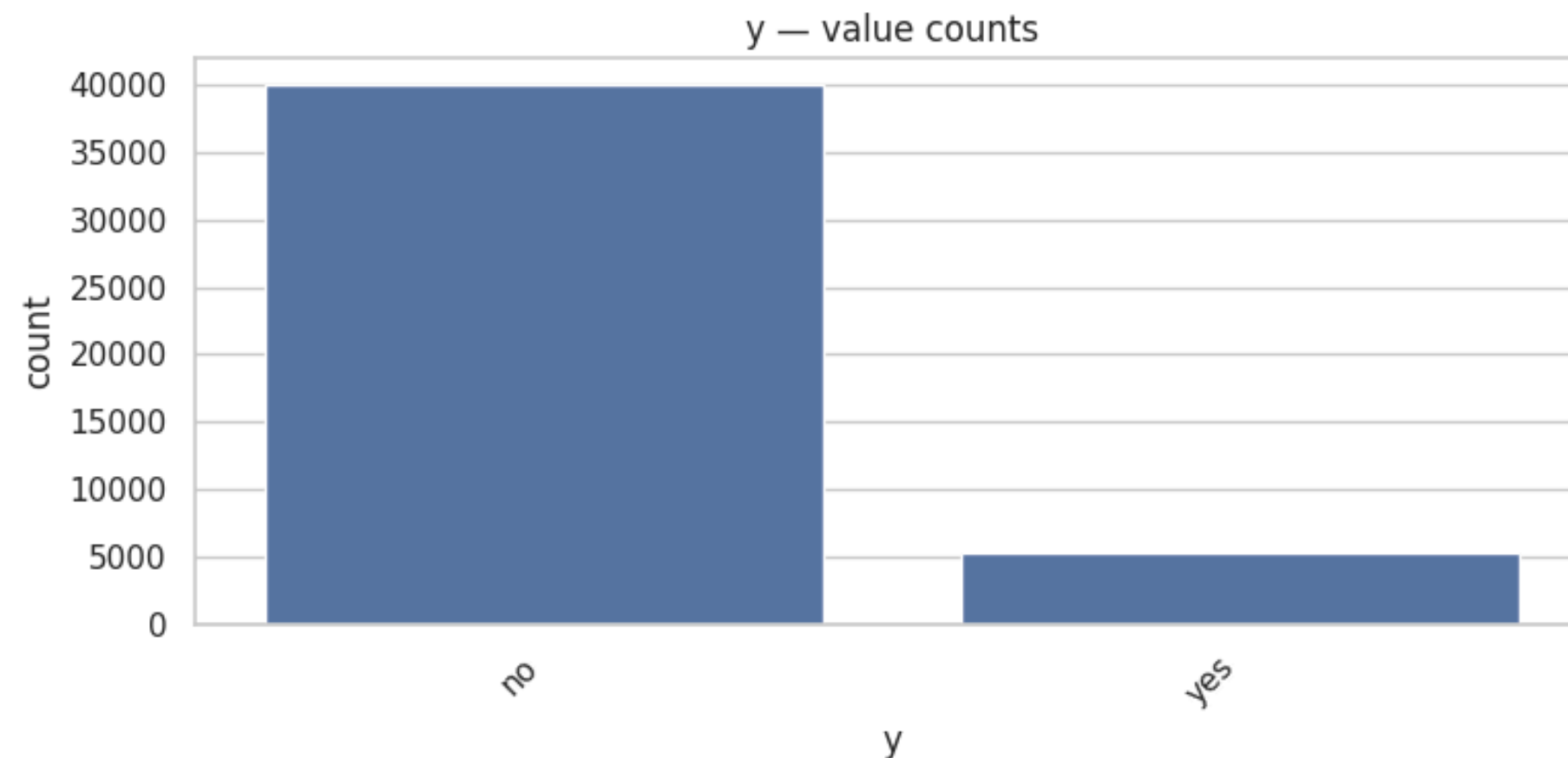
1. age (numeric)
2. job : type of job (categorical: "admin.", "unknown", "unemployed", "management", "housemaid", "entrepreneur", "student", "blue-collar", "self-employed", "retired", "technician", "services")
3. marital : marital status (categorical: "married", "divorced", "single"; note: "divorced" means divorced or widowed)
4. education (categorical: "unknown", "secondary", "primary", "tertiary")
5. default: has credit in default? (binary: "yes", "no")
6. balance: average yearly balance, in euros (numeric)
7. housing: has housing loan? (binary: "yes", "no")
8. loan: has personal loan? (binary: "yes", "no")
9. contact: contact communication type (categorical: "unknown", "telephone", "cellular")
10. day: last contact day of the month (numeric)
11. month: last contact month of year (categorical: "jan", "feb", "mar", ..., "nov", "dec")
12. duration: last contact duration, in seconds (numeric)
13. campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)
14. pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric, -1 means client was not previously contacted)
15. previous: number of contacts performed before this campaign and for this client (numeric)
16. poutcome: outcome of the previous marketing campaign (categorical: "unknown", "other", "failure", "success")
17. y - has the client subscribed a term deposit? (binary: "yes", "no")

Missing Attribute Values: None

Dataset Overview

This dataset consists of **45,211 rows** and **17 columns**

- **Bank client data:** age, job, marital, education, default, balance, housing, loan
- **Last-contact data:** contact, day, month, duration
- **Other campaign attributes:** campaign, pdays, previous, poutcome
- **Target:** y — did the client subscribe to a term deposit?
- "unknown" is a real category, not missing data: job (~0.6%), education (~4.1%), contact (~28.8%), poutcome (~81.7%)
- pdays == -1 is a sentinel meaning "never contacted before"
- Target y is imbalanced: ~11.7% yes / ~88.3% no





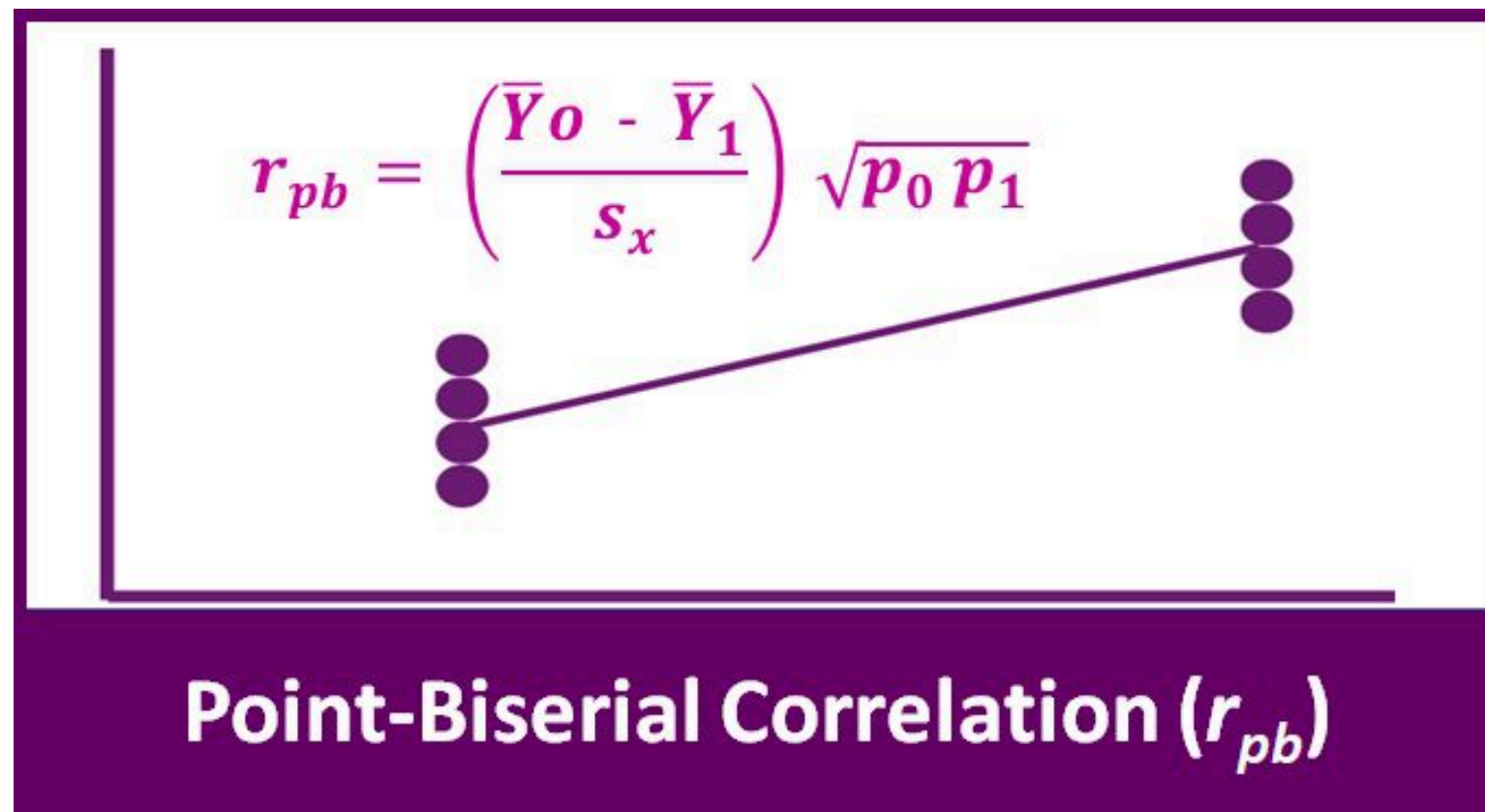
train.csv vs. test.csv

- **train.csv:** 45,211 rows, the full dataset, date-ordered May 2008 – Nov 2010
- **test.csv:** 4,521 rows — a random 10% subset drawn *from train.csv itself* (per UCI source docs), not an independent sample
- **Decision:** drop test.csv entirely; we run our own seeded train_test_split on train.csv at modeling time
- **Why:** using test.csv as a holdout risks leakage / duplicated rows since it isn't independent of train.csv

Classification EDA: Target y

Relevance Score

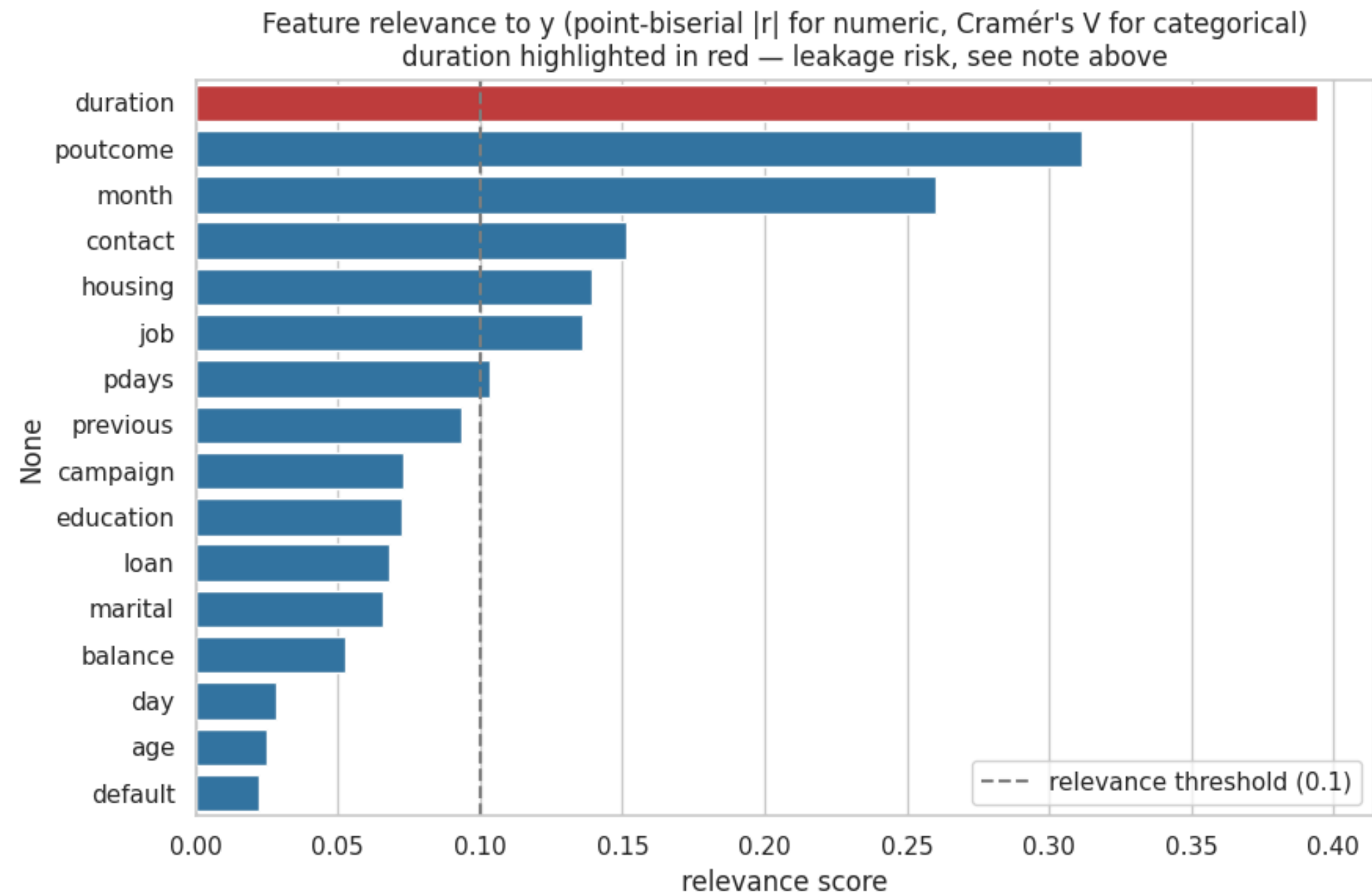
- 16 candidate features — screen with one ranked score before plotting anything
- **Numeric features** → **point-biserial correlation vs. y**
- **Categorical features** → **Cramér's V from the raw crosstab vs. y**
- **Point-biserial correlation**: same math as Pearson r, applied to a **continuous variable vs. a binary variable**. Range **-1 to +1** — sign = direction of the relationship, magnitude = strength.
- **Cramér's V**: built from a contingency table → chi-square test → normalized to **0–1**. 0 = no association, 1 = perfect association — no direction, unlike Pearson r. Caveat: inflated on small or sparse samples.



$$V = \sqrt{\frac{\chi^2}{N(k - 1)}}$$

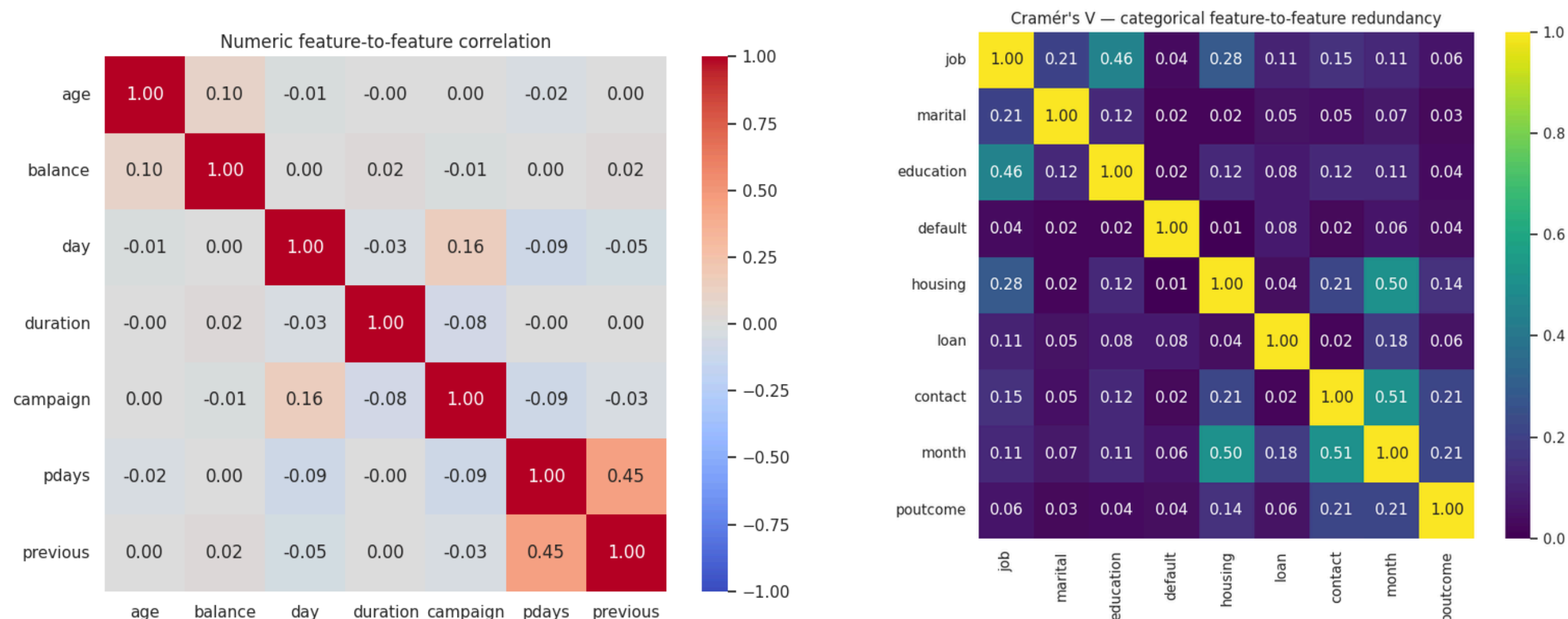
Relevance Score

- Threshold = 0.1 (Cohen's 1988 "small effect" convention)
- **7 features** clear it: **duration**, **poutcome**, **month**, **contact**, **housing**, **job**, **pdays**
- **duration (red)** is the top-ranked feature but is **leakage** - excluded from any model



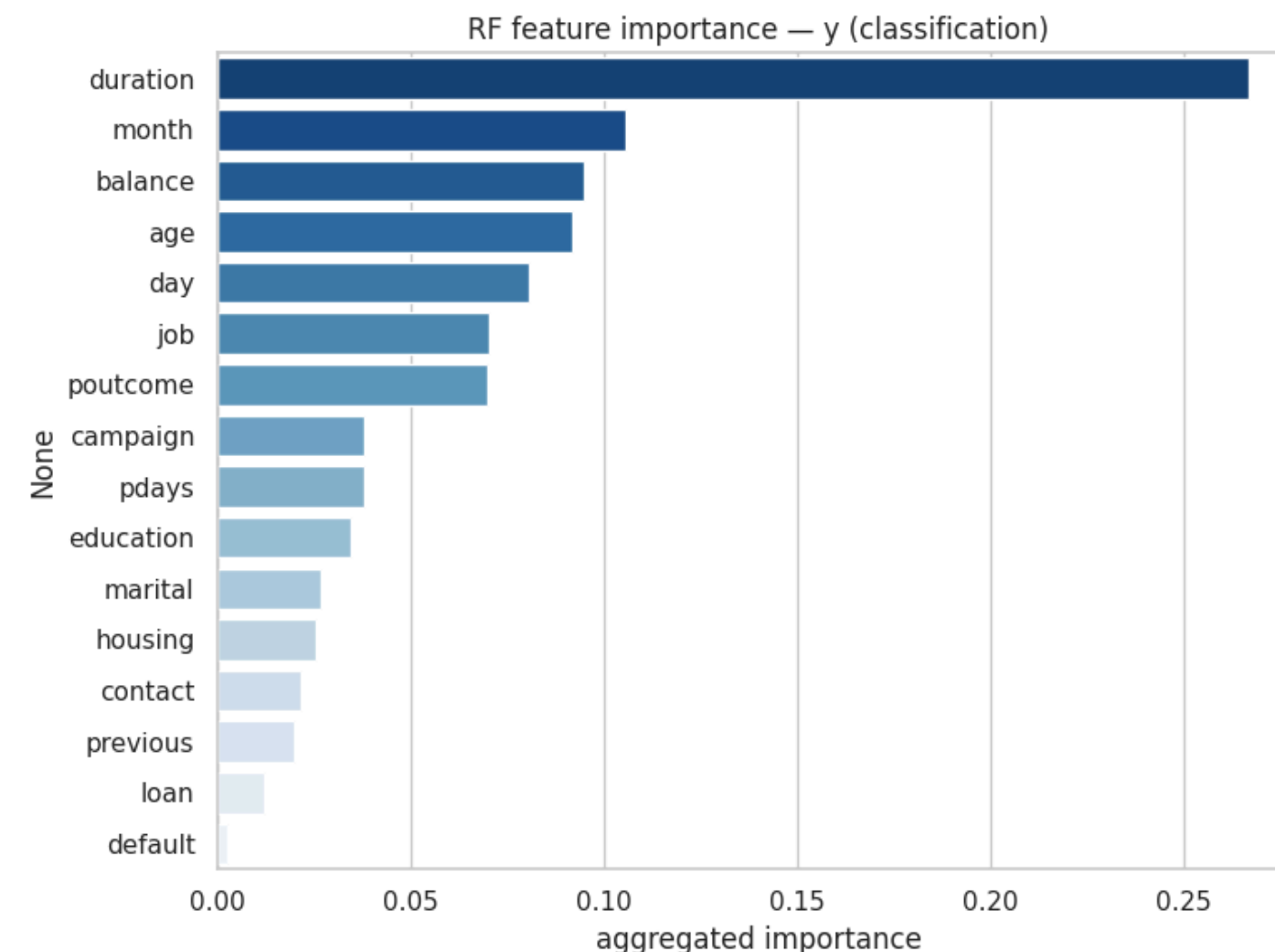
Feature vs Feature Correlation

- **Left chart — Pearson correlation** (numeric columns only: age, balance, day, duration, campaign, pdays, previous)
 - **Most numeric features are independent** — nearly every pair sits within ± 0.1
 - Two exceptions stand out: **pdays-previous** (0.45) and **day-campaign** (0.16)
- **Right chart — Cramér's V** (categorical columns only: job, marital, education, default, housing, loan, contact, month, poutcome)
 - **Most categorical features are independent** — most pairs sit below 0.15
 - Three pairs stand out: **contact-month** (0.51), **housing-month** (0.50), **job-education** (0.46)



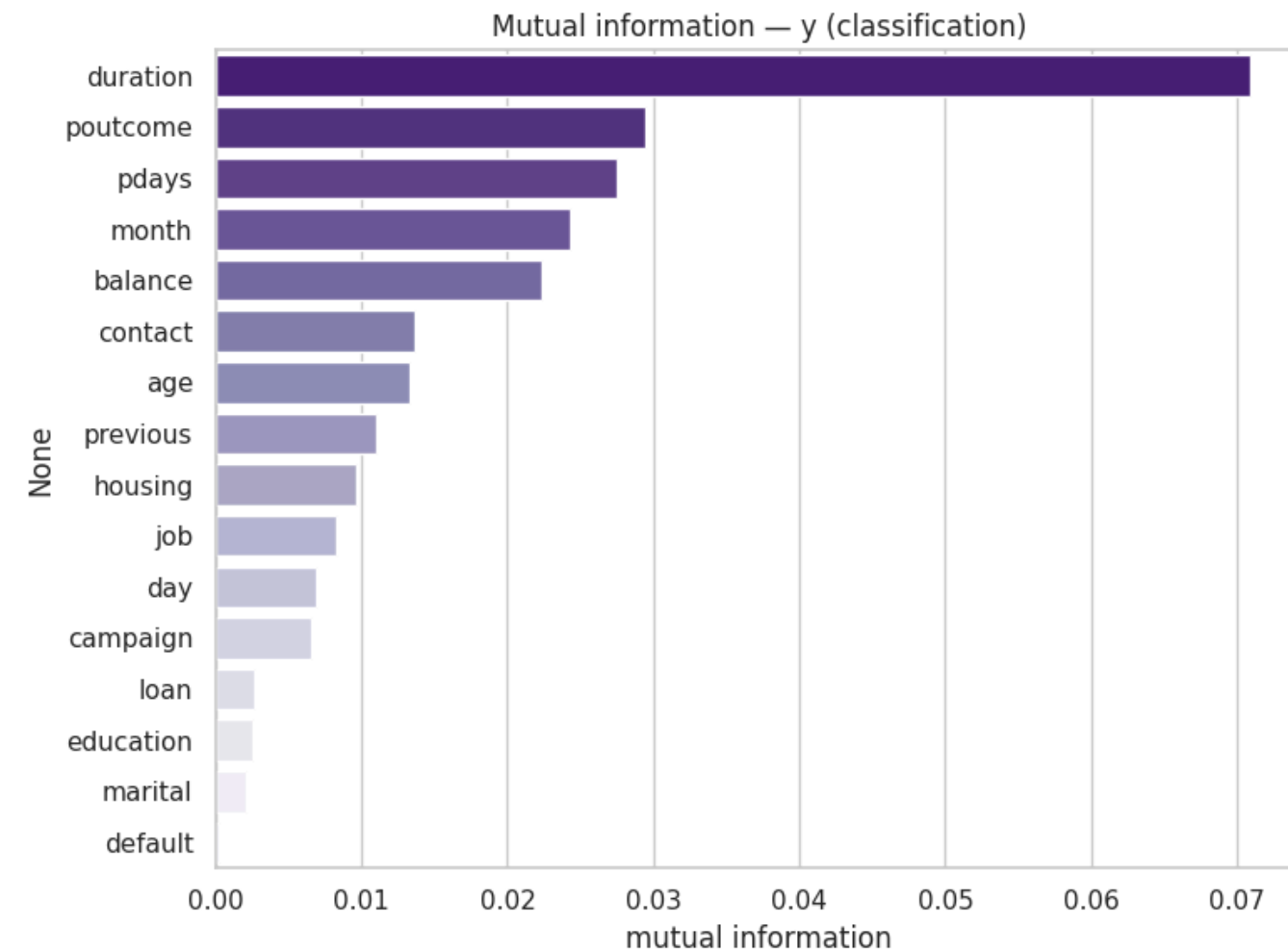
Random Forest Importance Score

- **Agrees with previous:**
 - duration, month, poutcome — independent confirmation from a completely different method
- **Disagrees with previous:**
 - **balance/age rank much higher here** (0.095/0.092) than in Step 1 (0.053/0.025) — real non-linear signal, or RF's continuous-feature bias? (flagged, not yet resolved)
 - **housing/contact rank lower here** — plausibly absorbed by month's redundant signal (see Step 2/3)



Mutual Information Cross-Check

- **Mutual Information:** measures shared info between a feature and y directly from the data. captures non-linear dependency (unlike Pearson/Cramér's V). Units: nats, so compare by rank, not raw value.
- **Top 5:** duration 0.0709 · poutcome 0.0294 · pdays 0.0275 · month 0.0244 · balance 0.0223
- **duration/poutcome/month** robust across all 3 methods; pdays jumps to 3rd — a genuine promotion, its "never contacted" structure was understated by linear correlation
- **balance/age** land in MI's top 7 too — independent, model-free confirmation this is real (if modest) signal, resolving the RF question from the last slide — but still tempers rather than overturns the original shortlist



Classification Summary

- **Features chosen:** poutcome, month, contact, housing, job, pdays
- **Why:** each cleared the Step 1 relevance threshold (0.1) and held up under two independent cross-checks (Random Forest, Mutual Information) — not a single method's quirk, agreement across 3 different kinds of test
- **duration** topped every method (correlation, RF, MI) but is **excluded** — it's only known after a call ends, so it can't be used to decide who to call before calling them
- **Class imbalance:** ~11.7% "yes" vs. ~88.3% "no" — needs class weighting, resampling, or threshold tuning; plain accuracy would be misleading

Regression EDA: Target `balance`

Relevance Score

- 16 candidate features — screen with one ranked score before plotting anything (y included as a candidate here too, mirroring how duration is scored, not skipped, in the classification track despite being unusable in a deployable model)
- **Numeric features** → **Pearson correlation vs. balance**
- **Categorical features** → **correlation ratio (η) from a one-way ANOVA vs. balance**
- **Pearson correlation**: same math as point-biserial in the classification track, applied to two continuous variables here instead of continuous-vs-binary. Range **-1 to +1** — sign = direction of the relationship, magnitude = strength.
- **Correlation ratio (η)**: built from a one-way ANOVA decomposition ($SS_{\text{between}} / SS_{\text{total}}$) → normalized to **0–1**. 0 = no association, 1 = category membership fully determines the group's average balance — no direction, unlike Pearson r. Caveat: inflated when a category has very few members (the η equivalent of Cramér's V's small-sample caveat).

Pearson Correlation Coefficient

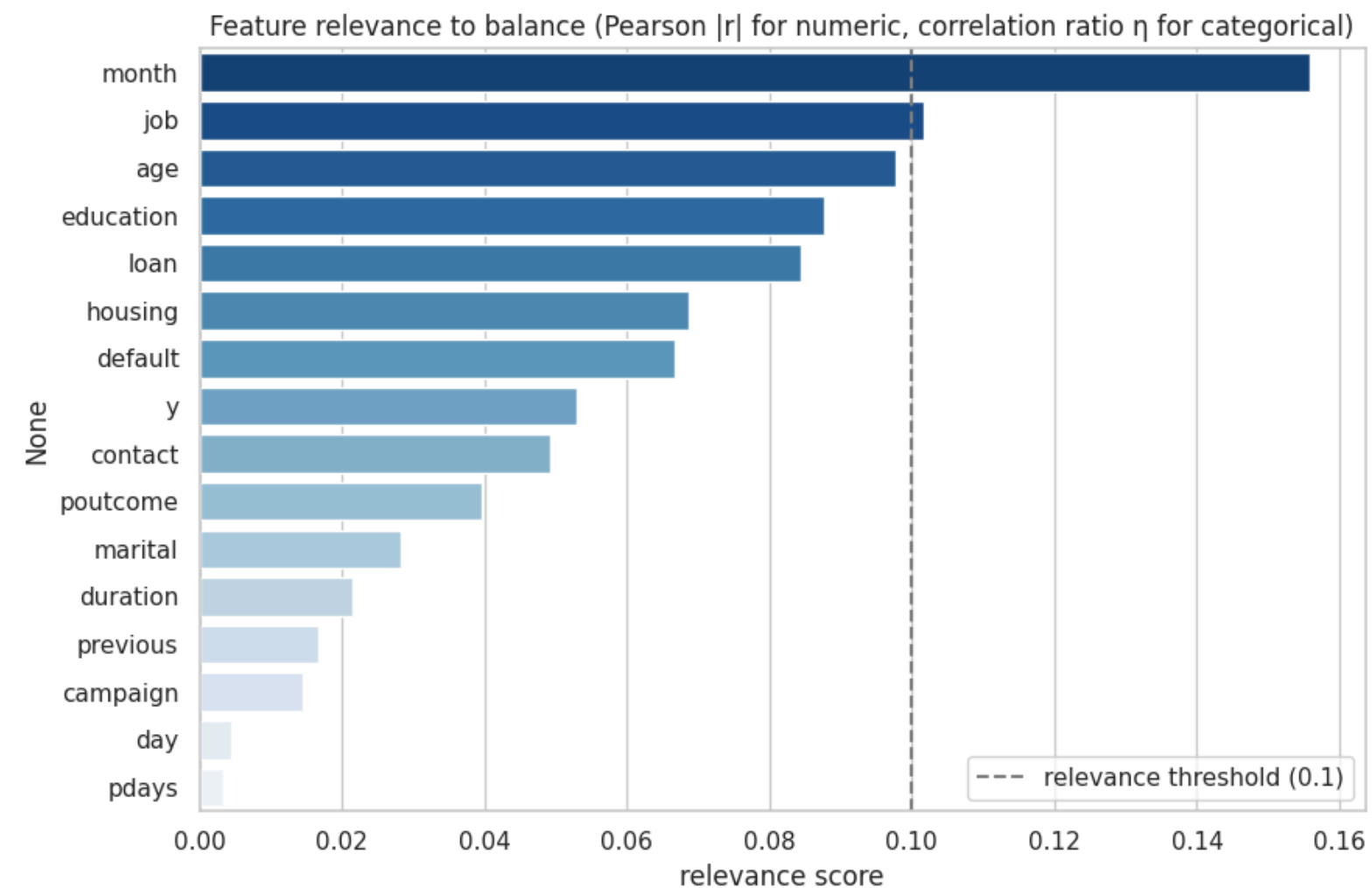
$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$



$$r = \frac{\sum z_x z_y}{n - 1}$$

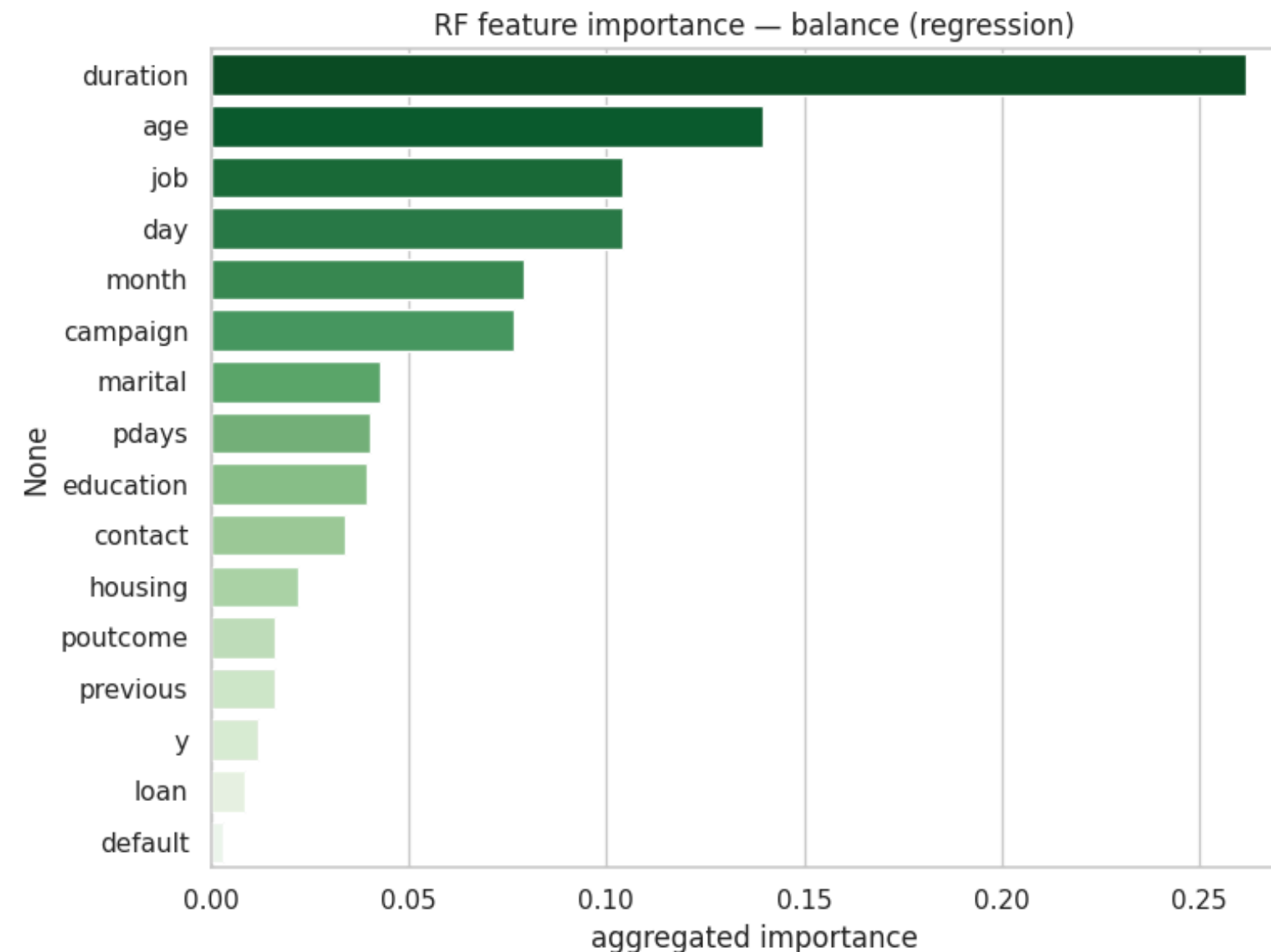
Relevance Score

- Threshold = 0.1 (Cohen's 1988 "small effect" convention)
- **2 features** clear it: **month, job**
- **age (0.098)** misses by a hair but isn't actually weak



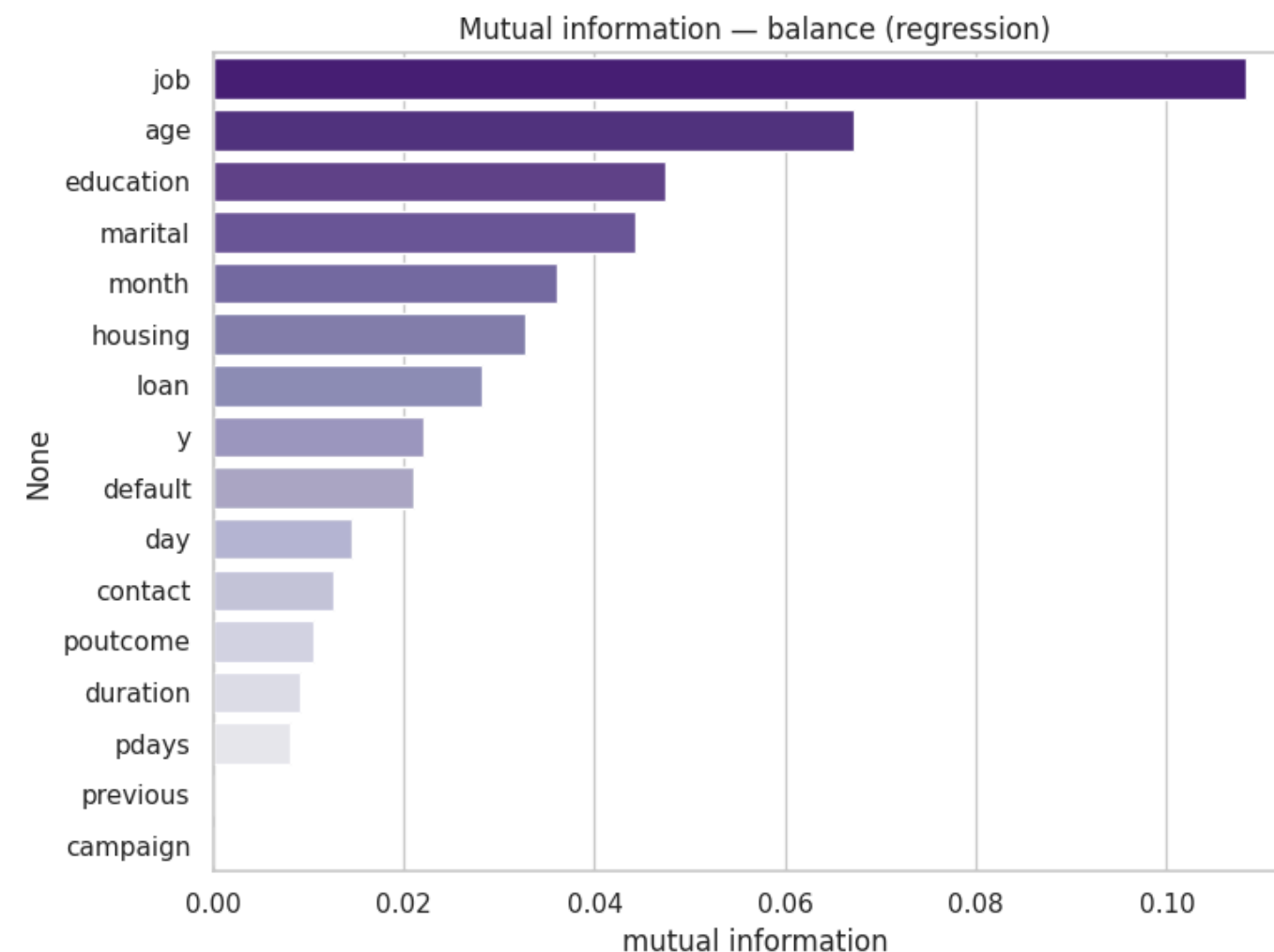
Random Forest Importance Score

- **Agrees with previous:**
 - **job** (independent confirmation, Step 1 2nd → RF 3rd), and age confirms the age-binned deep dive's finding that it carries real signal despite missing the Step 1 threshold
- **Disagrees with previous:**
 - **month** drops from 1st (Step 1) to 5th here — plausibly absorbed by its redundancy with contact/housing; duration's #1 RF rank contradicts its near-zero Step 1 score — flagged as bias, not yet resolved



Mutual Information Cross-Check

- **Mutual Information:** definition + non-linear/nats caveat
- **Top 5:** job · age · education · marital · month
- **job** robust across all 3 methods; marital is the new promotion story here (10th → 7th → 4th), the regression-track analog to pdays' jump in classification
- **duration** drops to 13th of 16 — this directly resolves the "not yet resolved" flag from the RF slide (C6): its #1 RF rank was an artifact, not real
- Kept the **age/month** and **y** lines as short trailers since they're still live threads from earlier slides, but led with your exact requested framing for the two headline findings



Regression Summary

- **Features chosen:** job, age (binned), month, housing, loan, education
- **No feature scores well in absolute terms, in any method:** best Step 1 score is month (0.156), best RF score is age (0.139), best MI score is job (0.108) — all still in Cohen's "small effect" range, nowhere near classification's duration(0.395)/poutcome(0.312). balance needs several modest signals combined, not one dominant driver.
- **Why these 6:** job/age/month rank consistently across all three methods even though their raw scores stay modest; housing/loan/education carry real signal too (clear boxplot median gaps, e.g. no-loan 496 vs. has-loan 258, and moderate MI ranks) but swing more across methods — most notably suppressed in RF, plausibly because month absorbs their shared signal (see C3, and the classification track's B5/B6)
- **duration** topped RF (0.262) but that rank isn't trustworthy — near-zero Step 1 score (0.022) and 13th-of-16 by MI both flag it as a continuous-feature artifact, not a real driver
- **y scores** real but modest everywhere (rank 8 in both Step 1 and MI) yet is excluded outright — same logic as duration in the classification track: it's the outcome of the same call a pre-call balance model would need to run before
- **Class imbalance:** balance is right-skewed with negative values present (min -8,019, max 102,127) — needs a signed-log or robust-scaling transform before modeling